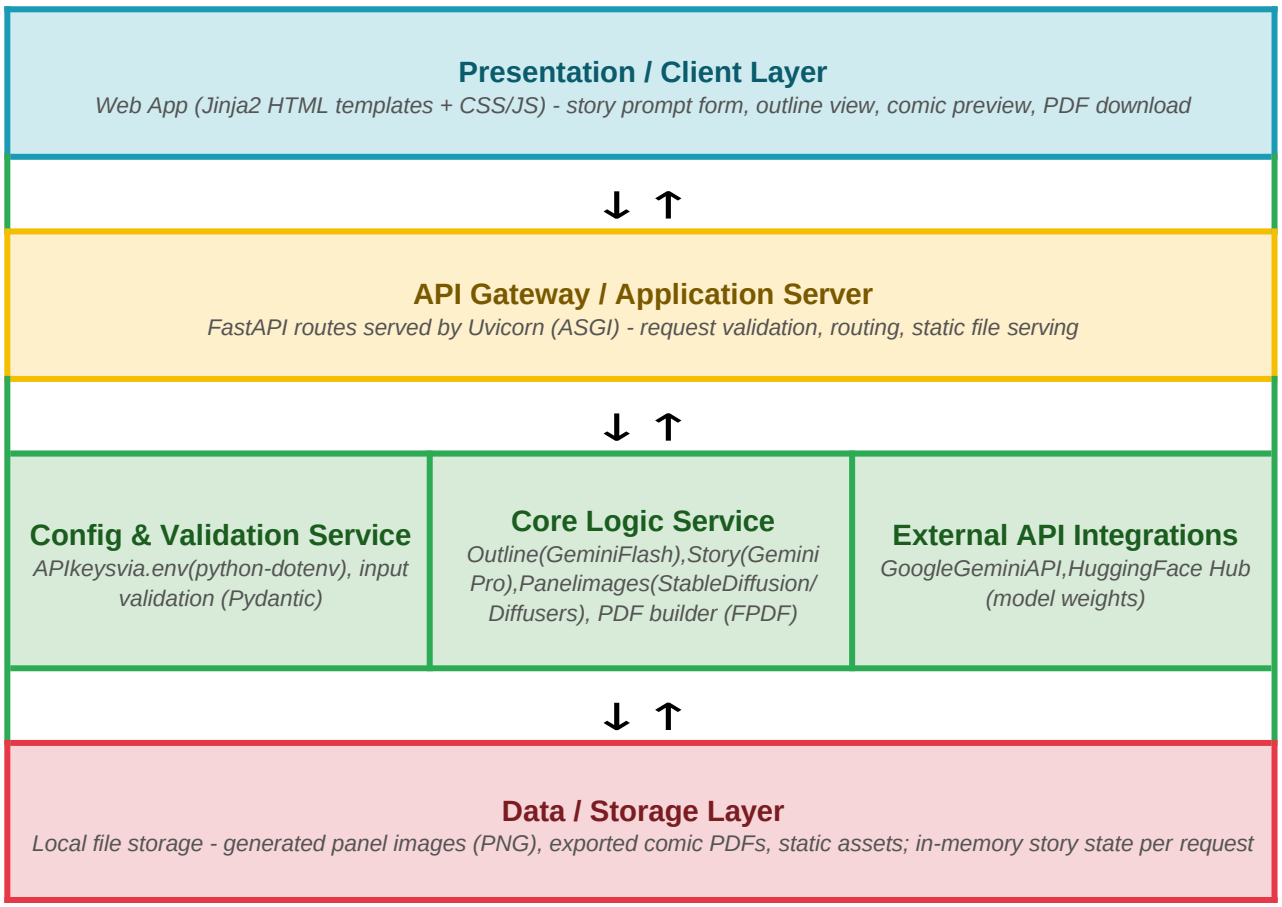
Solution Architecture

Date	3 October 2026
Team ID	SWTID-2026-9282
Project Name	ComicCraft - AI Comic Story Creator
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-levelviewoftheComicCraftarchitecture,mappingthePresentationLayer,ApplicationLayer(FastAPI gateway and AI services) and Data / Storage Layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Browser-based UI where the user enters a story idea, genre and number of panels, reviews the AI outlineandstory,viewsthegeneratedcomicpanels and downloads the PDF.	Jinja2templates,HTML5, CSS3, JavaScript
API Gateway	Single entry point that receives user requests, validates input, routes them to the correct service andservesgeneratedimagesandPDFsbacktothe client.	FastAPI,Uvicorn,Pydantic, python-dotenv
Microservices	Outline service (fast story skeleton), Story service (detailedscenes,dialogueandpaneldescriptions), Image service (one image per panel) and PDF service (compiles panels and text into a comic).	Gemini Flash, Gemini Pro (google-generativeai),Stable Diffusion with Diffusers + PyTorch, FPDF, Pillow
Database	Storesgeneratedpanelimagesandexportedcomic PDFs on disk so they can be previewed and downloaded;notrainingdatasetisneededbecause pre-trained models are used.	Local file system (static/ and output/folders),PNGandPDF files